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Task 5: Auto Tagging Support Tickets Using LLM

Objective

Automatically tag customer support tickets using an LLM (Groq + Llama-3.1).
Compare zero-shot (no examples) vs few-shot (with 4 examples) performance.

Dataset

Customer Support Ticket Dataset
Text = Ticket Subject + Description
Target = Ticket Type

Approach

- Model: llama-3.1-70b-versatile (fast, free tier)
- Sample: 100 tickets (quick run)
- Output: top category per ticket + simple accuracy

[ ]

```
!pip install groq pandas tqdm -q

import pandas as pd
from groq import Groq
from tqdm.notebook import tqdm

print("Libraries loaded")
```

```
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Libraries loaded
```

Variables Terminal

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```
print("Libraries loaded")
```

```
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Libraries loaded
```

```
GROQ_API_KEY = "gsk_............................" #dummy_key_for_demo_only
client = Groq(api_key=GROQ_API_KEY)
MODEL = "llama-3.1-70b-versatile"

print("Groq client ready")
```

```
Groq client ready
```

```
df = pd.read_csv("customer_support_tickets.csv")

df['text'] = (
    df['Ticket Subject'].fillna('') + " " +
    df['Ticket Description'].fillna('')
)

print("Dataset loaded")
print("Shape:", df.shape)
print("\nTicket Type distribution:")
print(df['Ticket Type'].value_counts())

n_samples = 100
df_sample = df.sample(n=n_samples, random_state=42).reset_index(drop=True)

print(f"\nSelected {n_samples} tickets for tagging")
display(df_sample[['text', 'Ticket Type']].head(5))
```

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```
display(df_sample[['text', 'Ticket Type']].head(5))
```

```
Dataset loaded
Shape: (14685, 18)

Ticket Type distribution:
Ticket Type
Technical issue         3029
Refund request          3020
Cancellation request    2932
Product inquiry         2857
Billing inquiry         2843
High                       1
2023-06-01 04:43:05        1
Name: count, dtype: int64

Selected 100 tickets for tagging
```

	text	Ticket Type
0	Battery life I'm having an issue with the {pro...	Billing inquiry
1	Network problem I've noticed a software bug in...	Billing inquiry
2	Hardware issue I'm having an issue with the {p...	Refund request
3	Data loss I'm having an issue with the {produc...	Cancellation request
4	Product recommendation I'm having an issue wit...	Cancellation request

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```
print("\nZero-shot results:")
display(df_sample[['text', 'Ticket Type', 'zero_shot_tag']].head(10))

zero_acc = (df_sample['Ticket Type'] == df_sample['zero_shot_tag']).mean()
print(f"Zero-shot accuracy: {zero_acc:.1%}")
```

```
Running zero-shot tagging...
Zero-shot: 100% ||||||
```

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```
few_acc = (df_sample['Ticket Type'] == df_sample['few_shot_tag']).mean()
print(f"Few-shot accuracy: {few_acc:.1%}")
print(f"Improvement: {few_acc - zero_acc:+.1%}")
```

```
Running few-shot tagging...
Few-shot: 100%|▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰▰| 100/100 [03:20<00:00, 2.70s/it]

Few-shot results:
```

	text	Ticket Type	zero_shot_tag	few_shot_tag
0	Battery life I'm having an issue with the {pro...	Billing inquiry	I would classify this support ticket as a **Te...	Technical Issue
1	Network problem I've noticed a software bug in...	Billing inquiry	The category for this support ticket is: **Tec...	Technical Issue
2	Hardware issue I'm having an issue with the {p...	Refund request	I would classify this support ticket as a **Te...	Technical Issue
3	Data loss I'm having an issue with the {produc...	Cancellation request	I would classify this support ticket as a **Te...	Technical Issue
4	Product recommendation I'm having an issue wit...	Cancellation request	I would classify this support ticket as a **Te...	Product Inquiry
5	Product setup I'm having an issue with the {pr...	Technical issue	I would classify this support ticket as a **Te...	Product Inquiry
6	Installation support I'm having an issue with ...	Refund request	I would classify this support ticket as a **Te...	Technical Issue
7	Installation support I'm having an issue with ...	Billing inquiry	I would classify this support ticket as a **Te...	Technical Issue
8	Account access I'm having an issue with the {p...	Cancellation request	I would classify this support ticket as a **Te...	Account Access
9	Delivery problem I'm having an issue with the ...	Cancellation request	I would classify this support ticket as a **Te...	Technical Issue

```
Few-shot accuracy: 5.0%
Improvement: +5.0%
```

Final Summary & Insights

Results (copy from output above):

- Zero-shot accuracy: 17.0%
- Few-shot accuracy: 5.0%
- Few-shot usually gives better results (examples teach the model)

Key takeaways

- LLMs can classify tickets quickly without training
- Prompt engineering (few-shot) improves accuracy noticeably
- Limitation: small sample — real system would use more data + better prompts

Skills demonstrated

- Prompt engineering (zero/few-shot)
- LLM API usage (Groq)
- Fast text classification evaluation

Security Note
Real API key removed before GitHub upload.

[ ]

```
df_sample.to_csv("task5_llm_tagging_results.csv", index=False)
print("Results saved → download task5_llm_tagging_results.csv")
```

```
Results saved → download task5_llm_tagging_results.csv
```